Ahan Gupta

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Github: https://github.com/spikerheado1234

Education

- **University of Illinois Urbana-Champaign** — Champaign, IL
  PhD in Computer Science - Advisor: Minjia Zhang — *Aug 2022 - Present*
- **National University of Singapore** — Singapore
  Bachelor of Computing in Computer Science — *Aug 2017 - Dec 2021*

Experience

- **Meta** — Menlo Park, CA
  Research Intern — *May 2026 - Current*
  - Writing high-performance GPU kernels for generative AI inference for Meta's frontier LLMs: Muse Spark.
- **Google DeepMind** — Mountain View, CA
  Student Researcher — *May 2024 - November 2024*
  - Investigated novel low-rank compression techniques to reduce KV-cache sizes for 10B+ parameter LLMs.
- **Citadel** — Hong Kong
  Software Engineering Intern — *May 2021 - Aug 2021*
  - Designed an authentication library to enable developers to integrate authentication logic with different services
  - Contributed to a tool that monitors AWS usage of different desks
  - Designed and built a monitoring tool that enables traders to track internal services' uptime and accuracy

Publications

- Davis Wertheimer*, Haochen Shen*, Ahan Gupta*, Derrick Liu, Yu Chin Fabian Lim, Mudhakar Srivatsa, Raghu K. Ganti, Minjia Zhang, Naigang Wang. A Self-Pruning Transformer: Extreme KV-Cache Compression with Universal Attention. COLM 2026.
- Muyan Hu, Ahan Gupta, Jiachen Yuan, Vima Gupta, Taeksang Kim, Xin Xu, Janardhan Kulkarn, Ofer Dekel, Vikram Adve, Charith Mendis. VTC: DNN Compilation with Virtual Tensors for Data Movement Elimination. OSDI 2026.
- Hoa La*, Ahan Gupta*, Alex Morehead, Jianlin Cheng, Minjia Zhang. MegaFold: System-Level Optimizations for Accelerating Protein Structure Prediction Models. ISC 2026. **Kernels adopted by AMD for OpenFold3**.
- Ahan Gupta*, Zhihao Wang*, Neel Dani, Masahiro Tanaka, Olatunji Ruwase, Minjia Zhang. AutoSP: Unlocking Long-Context LLM Training Via Compiler-Based Sequence Parallelism. ICLR 2026. **Adopted by Microsoft DeepSpeed**. **Featured by PyTorch Foundation blog**.
- Yueming Yuan, Ahan Gupta, Jianping Li, Sajal Dash, Feiyi Wang, Minjia Zhang. X-MoE: Enabling Scalable Training for Emerging Mixture-of-Experts Architectures on HPC Platforms. SC 2025. **Best Student Paper**.
- Ahan Gupta, Yueming Yuan, Devansh Jain, Yuhao Ge, David Aponte, Yanqi Zhou, Charith Mendis. SPLAT: Optimized GPU code generation framework for SParse reguLar ATtention. OOPSLA 2025.
- Ahan Gupta, Gefei Zuo, Aya Ibrahim, Adnan Aziz, Minjia Zhang. SketchLA: High-Performance Kernel Generation for Linear Attention through Reusable Program Sketches. In submission 2026.
- Ahan Gupta, Hao Guo, Yueming Yuan, Yanqi Zhou, Charith Mendis. FLuRKA: Fast fused Low-Rank & Kernel Attention. In Submission 2025. Preprint link: https://arxiv.org/abs/2306.15799.

* Denotes Equal Contribution

Service

- **ACM TACO Reviewer**: 2025, 2026
- **ISCA AEC**: 2024

Open Source Contributions

- **DeepSpeed**: (42k stars)
- **Enzyme**: (1.6k stars)